3D Vectors & the Cross Product

ANSWER KEY

Vector arithmetic in three dimensions

1 Given $\mathbf{a} = (2, -1, 3)$ and $\mathbf{b} = (1, 4, -2)$, find $\mathbf{a} + 2\mathbf{b}$.

$\mathbf{a} + 2\mathbf{b} = (2, -1, 3) + (2, 8, -4) = (4, 7, -1)$.

2 Find the magnitude of $\mathbf{v} = (6, -3, 2)$.

$|\mathbf{v}| = \sqrt{6^2 + (-3)^2 + 2^2} = \sqrt{36 + 9 + 4} = \sqrt{49} = 7$.

The scalar (dot) product in 3D

3 Find $\mathbf{a} \cdot \mathbf{b}$ for $\mathbf{a} = (3, 1, -2)$ and $\mathbf{b} = (-1, 4, 2)$, and use it to determine whether $\mathbf{a}$ and $\mathbf{b}$ are perpendicular.

$\mathbf{a} \cdot \mathbf{b} = 3(-1) + 1(4) + (-2)(2) = -3 + 4 - 4 = -3$. Since $\mathbf{a} \cdot \mathbf{b} \neq 0$, the vectors are not perpendicular.

4 Find the angle between $\mathbf{a} = (1, 0, 1)$ and $\mathbf{b} = (0, 1, 1)$, to the nearest degree.

$\mathbf{a} \cdot \mathbf{b} = 0 + 0 + 1 = 1$. $|\mathbf{a}| = \sqrt{2}$, $|\mathbf{b}| = \sqrt{2}$. $\cos\theta = \frac{1}{\sqrt{2} \cdot \sqrt{2}} = \frac{1}{2}$, so $\theta = 60^\circ$.

The vector (cross) product

5 Find $\mathbf{a} \times \mathbf{b}$ for $\mathbf{a} = (2, 1, -1)$ and $\mathbf{b} = (1, 3, 2)$, using $\mathbf{a} \times \mathbf{b} = (a_2 b_3 - a_3 b_2,\ a_3 b_1 - a_1 b_3,\ a_1 b_2 - a_2 b_1)$.

$a_2 b_3 - a_3 b_2 = 1(2) - (-1)(3) = 2 + 3 = 5$. $a_3 b_1 - a_1 b_3 = (-1)(1) - 2(2) = -1 - 4 = -5$. $a_1 b_2 - a_2 b_1 = 2(3) - 1(1) = 6 - 1 = 5$. So $\mathbf{a} \times \mathbf{b} = (5, -5, 5)$.

6 Explain why $\mathbf{a} \times \mathbf{b}$ is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$, and verify this for the result found in the previous problem by computing $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{a}$.

By definition, the cross product $\mathbf{a} \times \mathbf{b}$ produces a vector orthogonal to the plane containing $\mathbf{a}$ and $\mathbf{b}$, which is why it is perpendicular to both. Verifying: $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{a} = (5)(2) + (-5)(1) + (5)(-1) = 10 - 5 - 5 = 0$, confirming perpendicularity.

Area of a parallelogram and a triangle

7 Find the area of the parallelogram formed by $\mathbf{a} = (2, 1, -1)$ and $\mathbf{b} = (1, 3, 2)$, using the cross product found earlier ($\mathbf{a} \times \mathbf{b} = (5, -5, 5)$).

Area $= |\mathbf{a} \times \mathbf{b}| = \sqrt{5^2 + (-5)^2 + 5^2} = \sqrt{75} = 5\sqrt{3}$.

8 Triangle PQR has $\vec{PQ} = (3, 0, 4)$ and $\vec{PR} = (1, 2, 0)$. Find the area of the triangle.

$\vec{PQ} \times \vec{PR} = (0 \cdot 0 - 4 \cdot 2,\ 4 \cdot 1 - 3 \cdot 0,\ 3 \cdot 2 - 0 \cdot 1) = (-8, 4, 6)$. Magnitude: $\sqrt{64 + 16 + 36} = \sqrt{116} = 2\sqrt{29}$. Triangle area $= \frac{1}{2}|\vec{PQ} \times \vec{PR}| = \frac{1}{2}(2\sqrt{29}) = \sqrt{29}$.

Properties and applications of the cross product

9 Explain why $\mathbf{a} \times \mathbf{a} = \mathbf{0}$ for any vector $\mathbf{a}$, and why $\mathbf{a} \times \mathbf{b} = -(\mathbf{b} \times \mathbf{a})$ (the cross product is anti-commutative).

$\mathbf{a} \times \mathbf{a} = \mathbf{0}$ because the cross product's magnitude is $|\mathbf{a}||\mathbf{a}|\sin\theta$ where θ is the angle between the two vectors; since a vector makes an angle of 0 with itself, $\sin 0 = 0$, giving zero magnitude. The anti-commutative property $\mathbf{a} \times \mathbf{b} = -(\mathbf{b} \times \mathbf{a})$ follows directly from the determinant-style component formula: swapping $\mathbf{a}$ and $\mathbf{b}$ flips the sign of each component, reversing the direction of the resulting vector (by the right-hand rule, reversing the order of the two vectors reverses which way the perpendicular vector points).

10 Find a unit vector perpendicular to both $\mathbf{a} = (1, 2, 0)$ and $\mathbf{b} = (0, 1, 1)$.

$\mathbf{a} \times \mathbf{b} = (2 \cdot 1 - 0 \cdot 1,\ 0 \cdot 0 - 1 \cdot 1,\ 1 \cdot 1 - 2 \cdot 0) = (2, -1, 1)$. Magnitude: $\sqrt{4 + 1 + 1} = \sqrt{6}$. Unit vector: $\frac{1}{\sqrt{6}}(2, -1, 1)$.

The scalar triple product and volume

11 Find the volume of the parallelepiped formed by $\mathbf{a} = (1, 0, 0)$, $\mathbf{b} = (0, 2, 0)$, and $\mathbf{c} = (1, 1, 3)$, using the scalar triple product $V = |\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|$.

$\mathbf{b} \times \mathbf{c} = (2 \cdot 3 - 0 \cdot 1,\ 0 \cdot 1 - 0 \cdot 3,\ 0 \cdot 1 - 2 \cdot 1) = (6, 0, -2)$. $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = 1(6) + 0(0) + 0(-2) = 6$. Volume $= |6| = 6$.

12 Explain what it means, geometrically, if the scalar triple product $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$ equals zero, and give the name for three vectors satisfying this condition.

If the scalar triple product equals zero, the parallelepiped formed by the three vectors has zero volume, meaning the three vectors all lie in the same plane (they are linearly dependent). Vectors satisfying this condition are called coplanar.

Vector projections

13 Find the scalar projection of $\mathbf{a} = (3, 4, 0)$ onto $\mathbf{b} = (1, 0, 0)$, using $\text{proj} = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{b}|}$.

$\mathbf{a} \cdot \mathbf{b} = 3(1) + 4(0) + 0(0) = 3$. $|\mathbf{b}| = 1$. Scalar projection $= \frac{3}{1} = 3$.

14 Find the vector projection of $\mathbf{a} = (2, 3, -1)$ onto $\mathbf{b} = (1, 1, 1)$, using $\text{proj}_{\mathbf{b}}\mathbf{a} = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{b}|^2}\mathbf{b}$.

$\mathbf{a} \cdot \mathbf{b} = 2 + 3 - 1 = 4$. $|\mathbf{b}|^2 = 1 + 1 + 1 = 3$. Vector projection $= \frac{4}{3}(1, 1, 1) = \left(\frac{4}{3}, \frac{4}{3}, \frac{4}{3}\right)$.

Finding the angle of a parallelepiped's diagonal

15 A rectangular box has edges along the vectors $(4, 0, 0)$, $(0, 3, 0)$, and $(0, 0, 2)$ from one corner. Find the angle between the space diagonal of the box and the edge vector $(4, 0, 0)$, to the nearest degree.

The space diagonal is the sum of the three edge vectors: $\mathbf{d} = (4, 0, 0) + (0, 3, 0) + (0, 0, 2) = (4, 3, 2)$. $\mathbf{d} \cdot (4, 0, 0) = 16$. $|\mathbf{d}| = \sqrt{16 + 9 + 4} = \sqrt{29}$. $|(4, 0, 0)| = 4$. $\cos\theta = \frac{16}{4\sqrt{29}} = \frac{4}{\sqrt{29}} \approx 0.743$, so $\theta \approx 42^\circ$.

16 Given $\mathbf{a} = (1, 2, 2)$, verify that $|\mathbf{a}| = 3$, and hence find a unit vector in the same direction as $\mathbf{a}$.

$|\mathbf{a}| = \sqrt{1^2 + 2^2 + 2^2} = \sqrt{1 + 4 + 4} = \sqrt{9} = 3$, confirming the given magnitude. A unit vector in the same direction: $\hat{\mathbf{a}} = \frac{1}{3}(1, 2, 2) = \left(\frac{1}{3}, \frac{2}{3}, \frac{2}{3}\right)$.

Testing collinearity using vectors

17 Determine whether the points $A(1, 2, 3)$, $B(3, 6, 7)$, and $C(5, 10, 11)$ are collinear, by comparing the vectors $\vec{AB}$ and $\vec{AC}$.

$\vec{AB} = (3 - 1,\ 6 - 2,\ 7 - 3) = (2, 4, 4)$. $\vec{AC} = (5 - 1,\ 10 - 2,\ 11 - 3) = (4, 8, 8)$. Since $\vec{AC} = 2\vec{AB}$ (a scalar multiple), the two vectors are parallel and share the common point A, so A, B, and C are collinear.

18 Find the value of k such that the vectors $\mathbf{a} = (2, k, -1)$ and $\mathbf{b} = (4, -2, 3)$ are perpendicular.

For perpendicular vectors, $\mathbf{a} \cdot \mathbf{b} = 0$: $2(4) + k(-2) + (-1)(3) = 0$, so $8 - 2k - 3 = 0$, giving $5 = 2k$, so $k = \frac{5}{2}$.

19 A drone starts at position $(0, 0, 0)$ and flies with constant velocity vector $(3, 4, 2)$ metres per second. Find the drone's position after 5 seconds, and find the distance it has traveled from its starting point (assuming it flies in a straight line).

Position after 5 seconds: $(0, 0, 0) + 5(3, 4, 2) = (15, 20, 10)$. Distance traveled equals the magnitude of the displacement vector: $|(15, 20, 10)| = \sqrt{15^2 + 20^2 + 10^2} = \sqrt{225 + 400 + 100} = \sqrt{725} = 5\sqrt{29} \approx 26.9$ metres.

20 This question has three parts. A triangle has vertices $P(1, 1, 0)$, $Q(4, 1, 0)$, and $R(1, 5, 0)$. (a) Find the vectors $\vec{PQ}$ and $\vec{PR}$. (b) Find $\vec{PQ} \times \vec{PR}$. (c) Hence find the area of triangle PQR, and explain why the cross product points purely in the z-direction here.

(a) $\vec{PQ} = (4 - 1,\ 1 - 1,\ 0 - 0) = (3, 0, 0)$. $\vec{PR} = (1 - 1,\ 5 - 1,\ 0 - 0) = (0, 4, 0)$. (b) $\vec{PQ} \times \vec{PR} = (0 \cdot 0 - 0 \cdot 4,\ 0 \cdot 0 - 3 \cdot 0,\ 3 \cdot 4 - 0 \cdot 0) = (0, 0, 12)$. (c) Area $= \frac{1}{2}|\vec{PQ} \times \vec{PR}| = \frac{1}{2}(12) = 6$. The cross product points purely in the z-direction because all three vertices lie in the xy-plane (all have $z = 0$), so the triangle itself is flat in that plane, and a vector perpendicular to a plane parallel to xy must point along the z-axis.

21 This question has two parts. Forces $\mathbf{F}_1 = (2, -1, 3)$ N and $\mathbf{F}_2 = (-1, 4, 1)$ N act on an object at the point $(1, 0, 2)$, with position vector $\mathbf{r} = (1, 0, 2)$ measured from the origin. (a) Find the resultant force $\mathbf{F}_1 + \mathbf{F}_2$. (b) Find the moment (torque) of $\mathbf{F}_1$ about the origin, using $\mathbf{M} = \mathbf{r} \times \mathbf{F}_1$.

(a) $\mathbf{F}_1 + \mathbf{F}_2 = (2 - 1, -1 + 4, 3 + 1) = (1, 3, 4)$ N. (b) $\mathbf{M} = \mathbf{r} \times \mathbf{F}_1 = (1, 0, 2) \times (2, -1, 3) = (0 \cdot 3 - 2 \cdot (-1),\ 2 \cdot 2 - 1 \cdot 3,\ 1 \cdot (-1) - 0 \cdot 2) = (2, 1, -1)$ N·m.

22 Find the angle that the vector $\mathbf{v} = (1, -2, 2)$ makes with each of the three coordinate axes (its direction angles), to the nearest degree.

$|\mathbf{v}| = \sqrt{1 + 4 + 4} = 3$. The angle with the x-axis: $\cos\alpha = \frac{\mathbf{v} \cdot (1, 0, 0)}{|\mathbf{v}|} = \frac{1}{3}$, so $\alpha \approx 71^\circ$. The angle with the y-axis: $\cos\beta = \frac{-2}{3}$, so $\beta \approx 132^\circ$. The angle with the z-axis: $\cos\gamma = \frac{2}{3}$, so $\gamma \approx 48^\circ$.